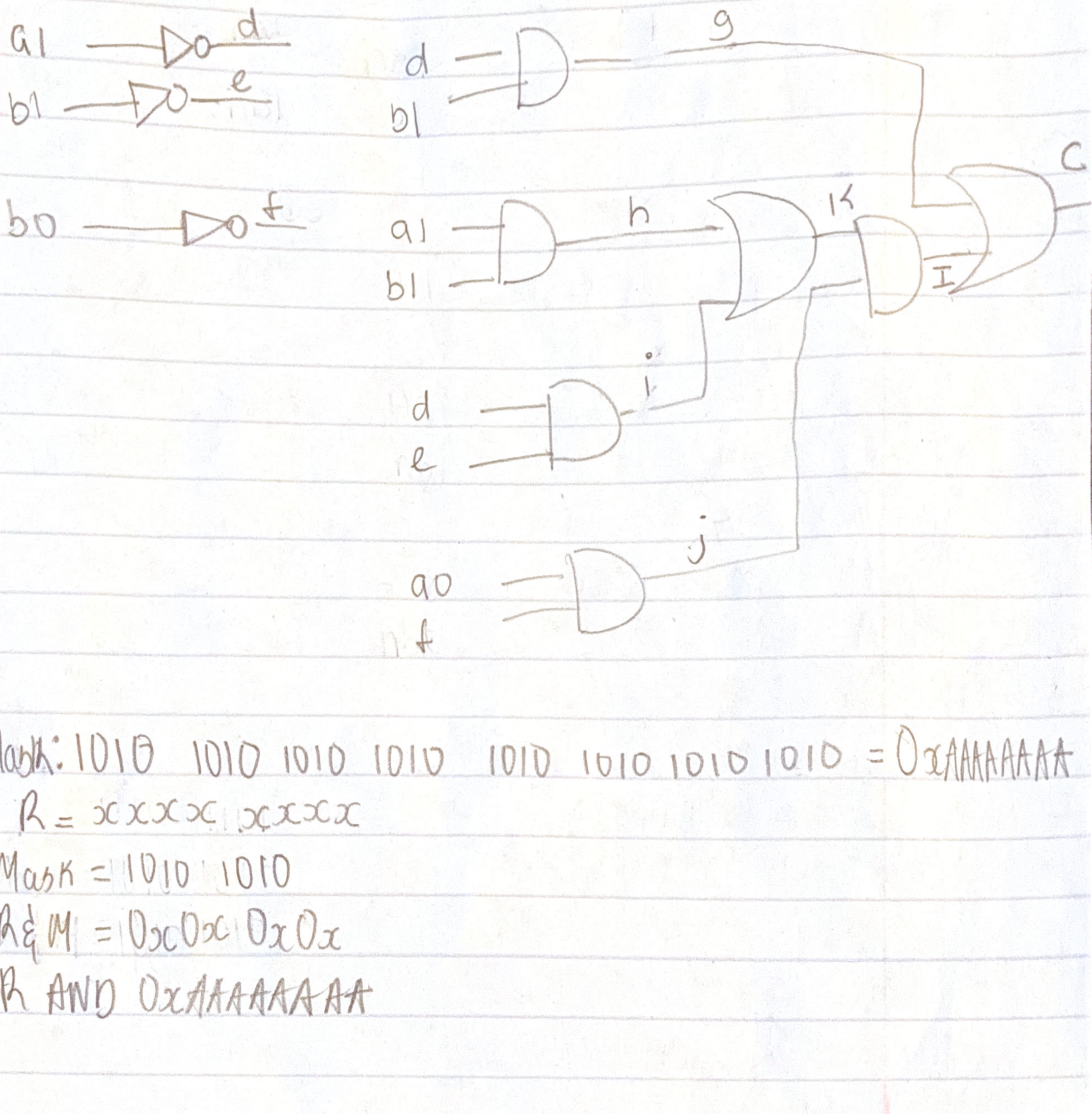
Homework 4.

1. Y: = (NOT A) OR C
   X: = NAND(NAND(A, A), NAND(B, C))

2. C = (Not a1 & b1) OR (a0 & Not b0 AND (a1 & b1) OR (NOT a1 & Not b1)))



3. Mask: 1010 1010 1010 1010 1010 1010 1010 1010 = 0xAAAAAAAA

a. R = xxxx xxxx
   Mask = 1010 1010
   R & M = 0x0x 0x0x
   R AND 0xAAAAAAAA

b. R = xxxx xxxx
Mask = 0000 0111
R OR M = xxxx x111
R OR 0x00000007

c. R = 1101 0110
Mask = 0000 0111
R & M = 0000 0110 = 2 + 4 = 6
R = R AND 0x00000007

d. R = xxxx xxxx xxxx
Mask = 1111 1111
R OR M = 1111 1111 = -1
R = R OR 0xFFFFFFFF

-128 64 32 16 8 4 2 1
1 0 0 1 0 1 1 0

e. R = 1001 0110
Mask = 1100 0000
R XOR M = 0101 0110
R = R XOR 0xC0000000

f. R = 1101 0110 128 + 64 + 16
Mask = 1111 1000
R AND M = 1101 0000 = 208
R = R AND 0xFFFFFFF8